

BrainCode: Agent-Created Language for Natural-Language Tasks and Conversations

Lia Soffer Noam Yehezkel
Tel Aviv University

Abstract

Formal intermediate languages could expose how large language models interpret natural language. We ask whether agents have sufficient understanding of linguistics to construct such a language, whether models interpret it consistently, and whether translation could improve reasoning. We introduce BrainCode, a typed language for tasks and conversations, intended for model comparison analyses, reasoning decomposition, and prompt formalization. A multi-model syntax loop and retrieval-assisted glossary swarm pursue coverage, determinism, expressivity, reasoning improvement, and human interpretability. Development across six datasets produces 657 active glossary records. Several models produced accepted translations for most sampled inputs, but success varied sharply across models. Qualitative analysis linked some failures to stricter interpretations of the glossary, highlighting the difficulty of making the language consistently usable across models. Symbol-use distributions and translation strictness differ across models, and round-trip reconstruction preserves broad structure but frequently loses task-specific details. Reasoning improvement experiment failed due to limited resources effect on design. These results suggest that BrainCode can expose model-specific representational preferences and behavioral differences when it comes to language perception, but does not yet demonstrate a model-neutral semantics or a reasoning benefit.

1 Introduction

Natural language permits multiple interpretations of tasks, conversations, and changing intentions. A shared intermediate representation could make these interpretations explicit by representing entities, actions, claims, relations, and conversational structure in a common notation. We investigate whether language-model agents can construct such a representation whether it generalizes to previously un-

seen inputs, and whether different models use it consistently.

Our work and objectives. BrainCode is a typed, human-readable intermediate language intended to encode the structure of tasks and conversations without simply copying their natural-language wording. It lies between natural language and a programming language: typed structure constrains composition, while many primitive meanings remain defined in prose. Possible uses are comparative research on model interpretations and linguistic tendencies, reasoning-task decomposition, and prompt formalization or compression. We operationalize five objectives. *Coverage* asks whether the frozen BrainCode specification and glossary can represent previously unseen evaluation items using definitions established during development. We operationalize this as the fraction of attempted translations accepted under the frozen language, including failures and errors in the denominator. This evaluates generalization to new items from known source distributions, rather than unseen domains. *Determinism* is stability of symbol-use distributions across repeated translations and models; lower divergence is better. *Expressivity* is preservation of source information through encoding and decoding, approximated by reconstruction similarity. *Improvement* is the accuracy difference between solving from a translated representation and solving from the original input. *Human interpretability* means that a reader can identify the roles, references, and intended meaning of statements from the specification and glossary. We maintain the last objective through readable names, examples, and human review; we do not quantitatively evaluate it or claim validated usability.

Research questions. **RQ1:** How stable or model-dependent is translation, and can its variation support comparative analysis of model behavior? We examine determinism alongside coverage. **RQ2:**

Can agents construct a usable language for heterogeneous natural-language material? We examine coverage, expressivity, and the interpretability-oriented design process. **RQ3:** Can an intermediate language improve reasoning through explicit decomposition? We compare direct and BrainCode-mediated solving. These exploratory questions concern one artifact; generating syntax need not establish semantic understanding.

2 Related work

Semantic and Intermediate Representations.

Abstract Meaning Representation (AMR; (Banasescu et al., 2013)) aim to encode sentence meaning independently of surface realization through a structured graph representation. BrainCode shares the goal of making semantic structure explicit, but also represents action plans, conversational turns, replies, and revisions, with a vocabulary expanded by agents. makes semantic consistency harder to guarantee: unlike a mature semantic formalism with a relatively fixed ontology, BrainCode allows many primitive meanings to remain defined in prose and introduces vocabulary through an iterative construction process.

Structured Reasoning and Programmatic Intermediate Representations.

Structured intermediate steps can help language-model reasoning when paired with suitable mechanisms. React (Yao et al. (2023)) interleaves reasoning traces with actions in an external environment, PAL (Gao et al. (2023)) delegates computation to executable programs, and parcel (Zelikman et al. (2023)) composes natural-language decompositions into programmatic structures. These approaches demonstrates that explicitly structuring intermediate reasoning can improve task performance when the representation is coupled to an execution mechanism or a task-specific decomposition procedure.

Formalization and Machine-Readable Representations.

Recent work on automatic formalization demonstrates the value of converting natural-language material into machine-checkable formal objects. For example, (Gloeckle et al., 2026) study textbook formalization into representations whose correctness can be mechanically checked. Our checker verifies selected language constraints, not proofs or semantic equivalence.

Emergent and Multi-Agent Communication

BrainCode is also related to work on emergent com-

munication between artificial agents. Prior studies have examined how agents can develop symbolic conventions or communication protocols under task pressure (Havrylov and Titov, 2017; Lazaridou and Baroni, 2020). BrainCode differs in two important respects. First, its communication system is intended to remain interpretable to humans rather than optimizing only task reward between agents. Second, its language is constructed through explicit proposal, critique, documentation, and human review rather than emerging solely from end-to-end optimization.

3 BrainCode

3.1 Syntax creation

The syntax loop assigns a Searcher to propose constructs, a Shaper to revise the specification, a Critic to test examples, and a Documenter to maintain definitions and glossary entries. The configuration uses Gemini 3.1 Pro Preview as Searcher, GPT-5.6 Terra as Critic, and Claude Sonnet 5 as Documenter; the Shaper rotates among them, with configured fallbacks. Starting from flat Python-like statements, explicit attribute/content distinctions, and conversation turns, prompts emphasize faithful representation, minimal primitives, explicit typing, bounded control flow, and readable names.

Proposals undergo simulated translations, critique, and documentation checks, with up to three reworks. Acceptance is qualitative. The resulting syntax uses typed tasks, actions distinct from checks, immutable bindings, bounded iteration, and explicit reply and revision links.

We subsequently compared and merged the syntax-loop product with Guy’s initial design template, adding the separation between descriptive TERMS, attributed CLAIMs, reasoning LINKs, and observed-event RECORDs while retaining the task and conversation structure. These distinctions support provenance and reduce confusion between describing and executing an action, at the cost of more categories and a larger interpretation burden. The migration therefore improves representational explicitness without establishing that models apply the distinctions consistently.

3.2 Glossary creation with a swarm

The adapted swarm infrastructure runs bounded batches of containerized translators, then migra-

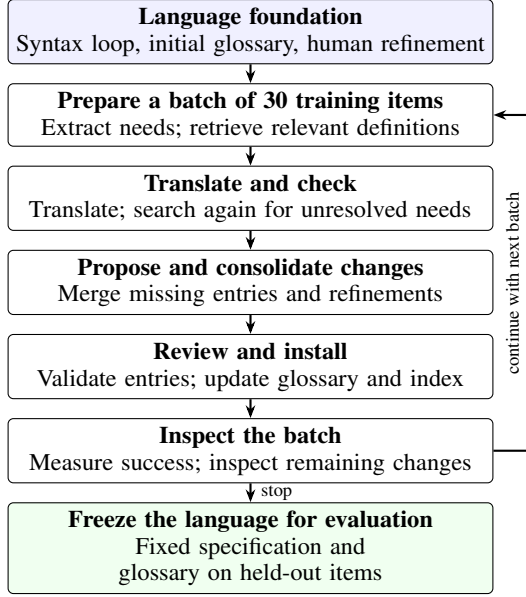


Figure 1: Construction loop and evaluation boundary. Human refinement supports development; each new batch uses the updated glossary. Held-out evaluation cannot modify the frozen language.

tion and inspection stages (Figure 1) until early stopping (batch with 100% successful translations, due to server rate limits). Humans still repaired runtime problems and made a small number of language corrections, so the final language reflects both agent proposals and human decisions. Humans also replaced enumerated nouns and values with lexical groups: `object_label::mug`, for example, uses a shared admission rule. Open label groups and standardized country/currency codes required corresponding signature, retrieval, and validation changes. This reduces enumeration; actions and relations still need explicit definitions, and label admission alone does not define semantic properties.

Data and loop. We assemble six datasets. Three contain relatively constrained tasks or action trajectories: Mind2Web for web interaction (Deng et al., 2023), ALFRED for household instructions (Shridhar et al., 2020), and the SWE-bench collection for software issues (Jimenez et al., 2024). Three provide more open-ended interactions: PRISM (Kirk et al., 2024), PATHs writing conversations (Mysore et al., 2025), and ThoughtTrace (Jin et al., 2026). This closed/open grouping does not imply a unique correct output for every task. Local 80/20 split, with ALFRED grouped by trial to avoid splitting descriptions of the same trial.

The planned development pool contains 500 train-

ing items per dataset, processed in batches of 30. Translators attempt faithful encodings and propose missing entries or refinements. Pre-migrators merge overlapping suggestions, a consolidator reconciles them, drafters prepare entries, and a stronger review stage can approve, replace, drop, or add proposals. The inspector stops when accepted edits are up to roughly 10% of the batch size (early stopping was defined due to server rate limitations).

Only 420 items in 14 batches were processed before stopping, rather than the planned 3,000. Figure 2 shows the initial burst of vocabulary requests and later residual gaps.

Retrieval and checking. Retrieval-augmented generation (RAG) limits BrainCode context usage to the compact specification and relevant glossary material. A needs extractor identifies the source meanings to represent. Segments of roughly 400 characters retain source references linking retrieval to the original need. Retrieval combines exact aliases with light stemming, BM25 lexical ranking, and multilingual MiniLM embeddings. It gathers 15 candidates per channel, combines rankings with reciprocal-rank fusion ($k = 60$), and reranks using 0.55 fusion score plus 0.45 context similarity. It retains 12 non-exact candidates plus exact matches, expands dependencies up to three levels and related entries by one hop, and includes shared rules. A widening operation increases candidates to 40 per channel and retains 30. Context used for reranking is capped at 600 characters. Group access is automatic: each context includes a group catalog, signatures identify admissible groups, and group-tagged needs retrieve exact group matches. New labels thus require no individual indexed entry.

Translators could retrieve, search, inspect entries, widen retrieval, and check outputs; migration rebuilds the in-memory index and embedding cache. The checker detects symbol occurrences, missing or unknown heads, invalid group atoms, retired entries, and opaque fallbacks; it is not a complete semantic validator.

3.3 Language and glossary

BrainCode combines a specification defining expression structure with a glossary defining vocabulary, together representing tasks, conversations, and attributed information. BrainCode distinguishes `MODE REQUEST`, which specifies intended work, from `MODE TRACE`, which records a conversation or trajectory. An `ENTRYPOINT` names a `TASK` or

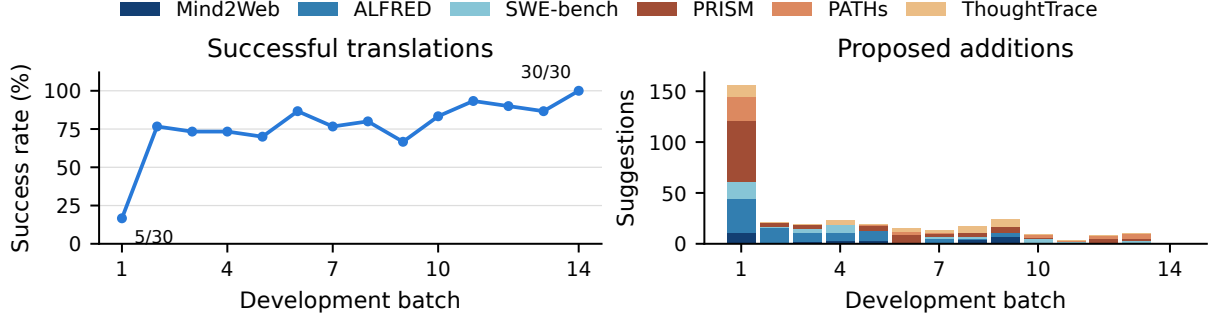


Figure 2: Development over 14 batches of 30 items. Left: successful translations as a percentage of each batch. Right: proposed glossary additions by dataset. Blue shades identify the three relatively closed-ended task datasets; warm shades identify open-ended dialogue datasets. Counts are proposals, not necessarily accepted entries. The changing glossary and batches preclude interpreting this as a held-out learning curve.

Kind	Count	Kind	Count
Value	315	Constructor	84
Claim relation	63	Structural	52
Operation	38	Composite	27
Category rule	27	Rule	14
Speech act	11	Attribute	8
Lexical group	8	Link	5
Example	5	Active total	657

Table 1: Active records in glossary version g19. Two deprecated composite records are excluded from the 659 stored records. Entries include rules and examples as well as symbols.

CONVO. Typed terms describe entities and activities; actions request effects, checks test conditions, and claims carry attribution or epistemic status. Named immutable bindings expose dependencies instead of hiding nested calls. Conditionals and bounded iteration structure tasks. Conversation turns identify speakers, support `REPLY_TO`, and distinguish whole-turn revisions from amendments to particular claims.

The glossary defines signatures and meanings (Table 1). Eight lexical groups avoid enumerating every possible value. Six permit normalized labels for objects, genres, foods, animals, colors, and platforms; two use country and currency standards. An atom such as `object_label::mug` preserves an object’s lexical identity, but does not supply an executable world model or a fully formal definition of a mug. This boundary between open labels and defined semantics becomes important in failures.

Translation examples. The task below represents “Move two pillows from the sofa onto the armchair,” adapted from the shipped examples with shorter local binding names. The action produces

references reused by the loop.

```
MODE REQUEST
ENTRYPOINT MovePillows
TASK MovePillows {
  ACTION pick_up(
    target=object_label::pillow, quantity
    =2,
    source=object_label::sofa)
  -> pillows : LIST[REF[STRING]]
  FOR EACH item IN pillows {
    ACTION place(target=item,
      destination=object_label::armchair,
      relation=on)
  }
}
```

The following short, constructed conversation illustrates a shared term and reply: user, “Shall we rinse the mug?”; agent, “I suggest rinsing it.” It demonstrates the notation rather than adding an evaluated success.

```
MODE TRACE
ENTRYPOINT Conversation
CONVO Conversation {
  TURN t1 SPEAKER=USER {
    TERM activity(verb="rinse",
      object=object_label::mug) -> a :
    TERM
    UTTER ask(target=a)
  }
  TURN t2 SPEAKER=AGENT REPLY_TO t1 {
    UTTER propose(target=t1.a)
  }
}
```

The activity is described, not executed. Both examples remain partly dependent on a reader’s interpretation of documented primitives.

4 Evaluations

4.1 Design and coverage (RQ1–RQ2)

We sample eight local test items per dataset (48 total), 3 times each for Gemini Flash 3.7 regular

and high-effort (288 translations total), In addition 3 items per item for Claude Sonnet 5.5, Claude Haiku 5.4, GPT 6.1 Sol, GPT 6 Luna, GPT o4-mini form a shared 18-item subset (270 translations total). All conditions share the translator toolkit and precomputed item needs and retrieval, controlling initial evidence but not subsequent tool use or rule interpretation. We retain infrastructure errors in coverage denominators and distinguish explicit translation failure from errors or rejected successes.

Haiku has the highest shared coverage, followed by high-effort Flash, Flash, and Sonnet. Therefore the data do not support a blanket advantage of Gemini over Claude, nor a monotonic relationship with nominal model strength. The per-dataset profiles in Figure 3 show additional variation: Sonnet accepts relatively more web tasks than PATHs conversations, while Haiku maintains high coverage across sources.

Condition	S/F/E	Cov.%	Miss.	All%
Flash high	49/2/3	90.7	1.0	91.0
Flash	46/4/4	85.2	1.8	88.9
Sonnet 5.5	24/30/0	44.4	1.8	—
Haiku 4.5	50/4/0	92.6	2.5	—
o4-mini	4/46/4	7.4	2.4	—
GPT-6.1 Sol*	0/19/4	0.0	—	—
GPT-6 Luna*	0/19/3	0.0	—	—

Table 2: Shared-set coverage: successes/failures/errors (S/F/E) out of 54 attempts; Miss. is mean reported missing symbols per failed attempt. All is coverage over 144 Gemini attempts. *Incomplete diagnostic samples of 23 and 22 attempts, respectively, not balanced comparisons.

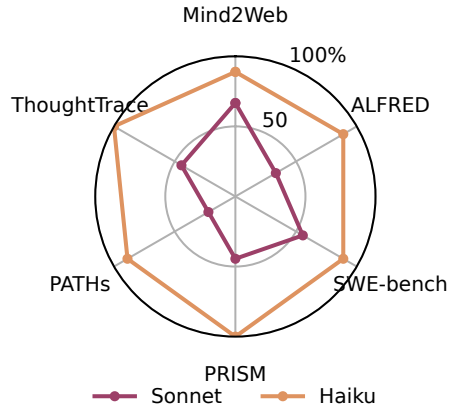
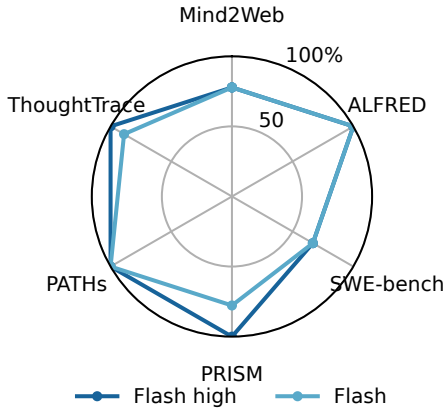


Figure 3: Shared-set coverage by dataset: nine attempts (three items, three repeats) per model and axis. The two panels have identical scales. Variation reflects both item difficulty and model-specific interpretation; each axis is based on only three distinct items.

Why OpenAI trials failed. GPT-6.1 Sol and GPT-6 Luna had no accepted translations in incomplete trials. Their stricter readings requested dedicated constructors where other models used general ones, alongside genuine glossary gaps. o4-mini succeeded in 4/54 attempts; some failures used prohibited prose fallbacks. These conditions are excluded from main comparisons; o4-mini remains a failure-confounded diagnostic. Interpretation strictness and Gemini-led construction may both contribute.

4.2 Determinism (RQ1)

For each translation, we count recognized glossary-symbol occurrences and lexical-group atoms, excluding quoted literals, comments, and structural/rule/example tokens. We normalize counts into distributions. For distributions P, Q over their union of symbols, with $M = (P + Q)/2$, we use base-2 Jensen–Shannon divergence:

$$JS(P, Q) = \frac{1}{2} D_{KL}(P \| M) + \frac{1}{2} D_{KL}(Q \| M). \quad (1)$$

Between-model JS compares counts pooled over the shared sample. Self-divergence averages all repeat-pair divergences within each item, then averages items with at least two usable runs. We use the implemented arithmetic means; an earlier design proposed harmonic averaging.

Figure 4 shows much closer pooled symbol use between Flash conditions (0.020) than across Gemini and Claude (0.094–0.102); Sonnet–Haiku divergence is 0.069. Mean self-divergence is 0.092 for high-effort Flash, 0.114 for Flash, 0.121 for Sonnet, and 0.173 for Haiku. High coverage and repeat stability thus need not coincide: Haiku covers more in-

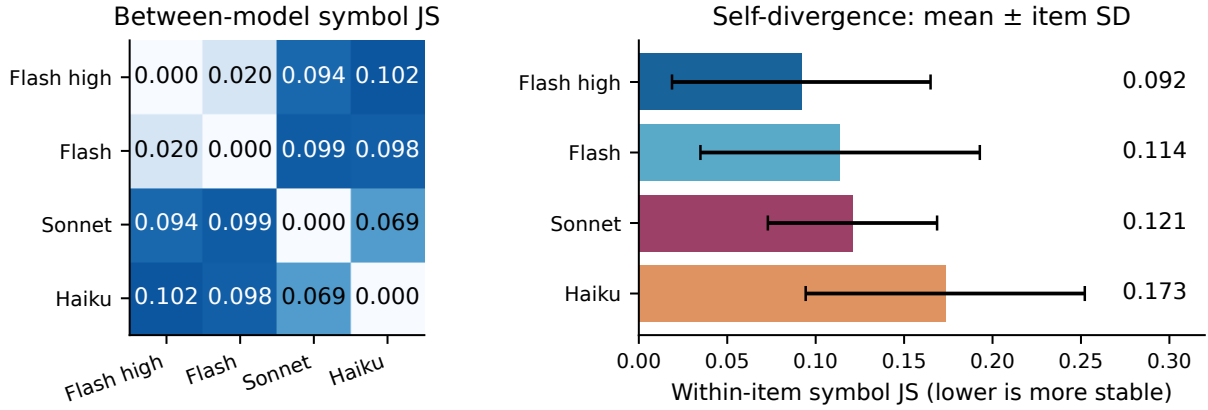


Figure 4: Determinism on the shared sample. Left: JS between pooled symbol distributions. Right: mean within-item repeat divergence, with standard deviation across items (not confidence intervals). Flash conditions have 17 eligible items each; Sonnet and Haiku have 18. Lower values indicate more similar symbol usage.

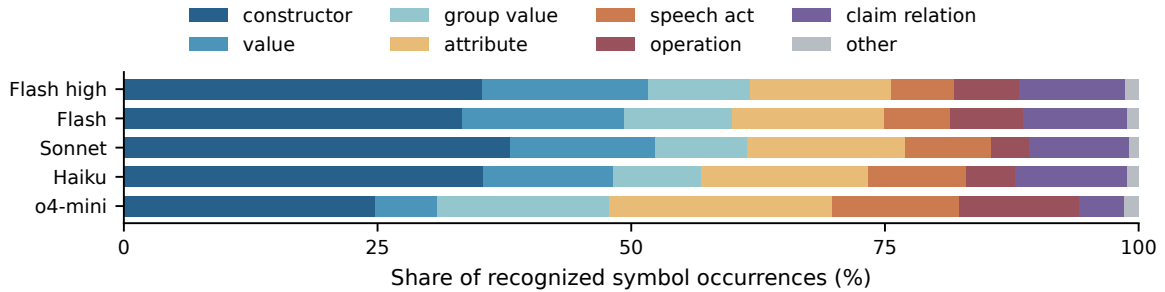


Figure 5: Normalized type mix of recognized symbol occurrences on the shared set, including explicit failures. Four main conditions have similar broad profiles despite different coverage. o4-mini is shown diagnostically; its 7.4% coverage makes a direct comparison of successful language use inappropriate. “Other” combines links and composites; group values are lexical-group atoms.

puts but varies more between runs. Neither pooled similarity nor self-stability demonstrates fidelity to the source.

Broad type distributions are closer than individual symbol distributions: main-condition pairwise type JS ranges from 0.0007 to 0.0071. Figure 5 gives no simple ordering of type preferences by success rate. o4-mini differs more (type JS 0.050–0.062 versus the other conditions), but its partial failures confound interpretation. Gemini seems to generate more action/task-oriented encodings, while Claude tends to generate more explicit utterance-oriented encodings, with Sonnet often constructing more terms and Haiku varying its decomposition. Selected high-divergence cases explain patterns rather than estimate their prevalence. Together with the coverage radar, they reveal representational preferences, without validating a taxonomy of linguistic abilities.

4.3 Expressivity (RQ2)

We encode natural language into BrainCode and have the same model decode it using the specification and used definitions, without the original text. Taking the first usable encoding among three repeats yields 46 reconstructions per Gemini condition: 45 Flash and 43 high-effort encodings were accepted; the remainder were explicit failures.

The Hebrew control uses the same model and harness for source-to-Hebrew translation and fresh-context reconstruction, withholding the original and omitting BrainCode context. Code, paths, identifiers, UI labels, URLs, and names may remain verbatim. Of 48 Hebrew round trips per condition, comparisons use the 46 paired BrainCode items.

Both routes use mean sentence BLEU with exponential smoothing (Papineni et al., 2002), ROUGE-L F1 (Lin, 2004), and word edit similarity $1 - d(x, \hat{x}) / \max(|x|, |\hat{x}|)$, where d is unit-cost word Levenshtein distance. Text is lowercased and conversational markers are removed. These metrics

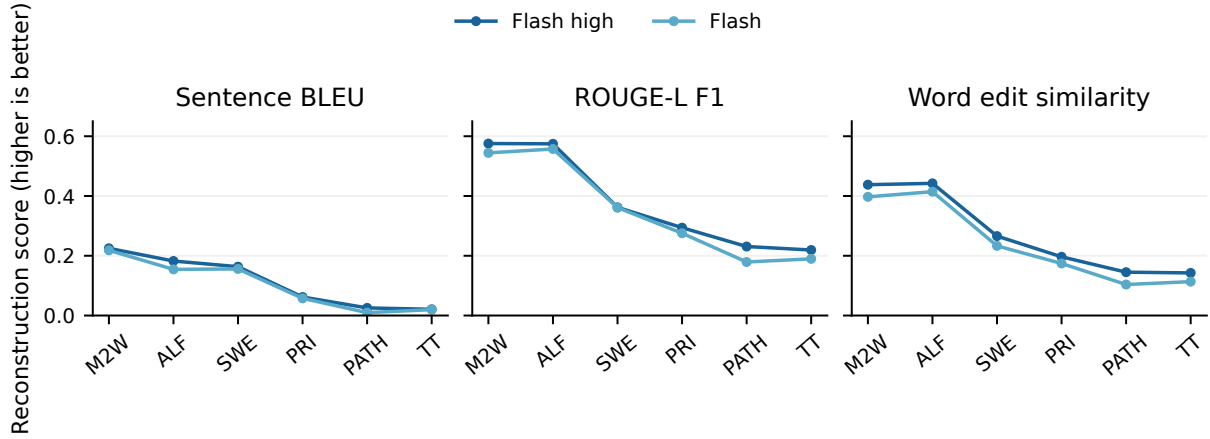


Figure 6: Mean round-trip scores by dataset for 46 reconstructions per Gemini condition (7–8 per dataset). M2W: Mind2Web; ALF: ALFRED; SWE: SWE-bench; PRI: PRISM; PATH: PATHs; TT: ThoughtTrace. Higher is better. Scores reflect both encoder and decoder behavior and include a small number of failed encodings.

Model	Route	BLEU	R-L	Edit
Flash	BrainCode	.103	.355	.242
Flash	Hebrew	.770	.893	.860
Flash high	BrainCode	.114	.380	.275
Flash high	Hebrew	.781	.898	.866

Table 3: Paired round-trip comparison on 46 items per model. R-L: ROUGE-L F1; Edit: word edit similarity. All scores range from 0 to 1; higher is better.

measure lexical reconstruction, not semantic equivalence.

Hebrew leads on every aggregate metric and dataset (Table 3), with higher BLEU on all paired items. Thus, a second translation step need not cause such severe reconstruction loss. Plausible contributors include model familiarity with Hebrew, BrainCode’s restricted vocabulary and decomposition requirements, and preserved identifiers benefiting Hebrew on technical tasks.

BrainCode better reconstructs constrained instructions than dialogue (Figure 6). Qualitative examinations revealed compression, step merging/splitting, and lost details: Flash drops an ALFRED book title retained by high-effort Flash; a PATHs height loses its value; French ThoughtTrace returns largely in English. Yet rewriting a third as approximately 33.33% need not lose information: wording differences are not necessarily semantic failures.

4.4 Reasoning improvement (RQ3)

We tested bidirectional translation fine-tuning of Qwen2.5-7B-Instruct on ProofWriter. Both lowered accuracy; Qwen also declined on English in-

puts after adaptation. Appendix C details the experiments, training, and incomplete verdict-decoding stage. Translation loss, extra context, limited data, and training to translate rather than reason plausibly contribute. Adaptation builds on pretrained representations (Radford et al., 2018), with overfitting and forgetting risks (Howard and Ruder, 2018); these mechanisms and general capability loss are not established here.

5 Conclusions and Discussion

5.1 Conclusions

Agent-created language (RQ2). Agents and human reviewers produced a substantial specification and glossary designed for readability; several models accepted most held-out translations. Yet noun grouping remained a recurring development and evaluation bottleneck: open object labels must support new values without silently inventing semantics, while closed definitions demand continuing enumeration. Failure reports identify this as a central problem without quantifying its causal share. High acceptance also coexists with lossy reconstructions: the paired Hebrew control preserves substantially more wording across all six datasets. This gap shows that coverage alone is insufficient evidence of a well-functioning language and motivates direct evaluation of which meanings survive translation.

Haiku exceeds Sonnet despite their expected capability ordering; Gemini exceeds Sonnet and the OpenAI conditions, but not Haiku. This suggests adaptation to the Gemini-led swarm (assuming Haiku 4.5 resembles Gemini Flash 3.7 more than

Sonnet 5.5), or a division between lenient readers (Gemini and Haiku) and stricter readers demanding dedicated constructions. Mixing formal constraints with prose meanings leaves difficult interpretive ambiguities that our process did not resolve across model families.

Comparative research (RQ1). Symbol distributions reveal decomposition and vocabulary preferences, making BrainCode a promising behavioral probe. It is not a ready-to-use comparative method: construction-model bias, failures, and checker strictness can confound linguistic tendencies. Repeated construction and independent semantic judgments must establish whether these preferences are robust.

Reasoning (RQ3). Neither in-context use nor our translation fine-tuning improved reasoning. Qwen’s English-input accuracy also decreased after adaptation, so translation quality alone cannot explain the observed deterioration. Limited, task-mismatched training and interference with prior behavior are plausible explanations, not established causes. Further assessment must separate representation quality, language fluency, reasoning training, and prompting costs.

5.2 Limitations

Our six sources cover a limited set of natural-language domains, with only 18 shared translation-evaluation items, 50 BBEH-mini reasoning items for Llama, and 70 ProofWriter items for Qwen (67 successfully translated). Repeated generations characterize decoding variability but do not constitute independent tasks. We also do not isolate the effects of loop design, retrieval, consolidation, prompting choices or human intervention.

Proxy request limits constrained swarm development, limited data extent in training and test sample sizes, and required interventions, including prompt adaptations and changes to command access during runs. Their indirect effects remain unmeasured: a frozen evaluation snapshot does not make development a controlled ablation. Model labels and proxy endpoints also limit reproducibility of hosted-model behavior.

The acceptance gate is incomplete, reconstruction metrics are lexical rather than direct measures of semantic equivalence, and interpretability lacks a human study. Runtime and token efficiency were excluded as objectives; the additional translation call and documentation imply overhead, but we did

not establish an efficiency benefit. Finally, bias toward models involved in language construction limits the fairness of using BrainCode to compare other model families.

Because BrainCode is a newly constructed language, there are no established standard metrics for its coverage, determinism, or expressivity, so these evaluations are mostly descriptive and rely on project-specific operationalizations.

5.3 Future work

With more resources, we might improve the experimental pipeline to increase potential BrainCode performance: a more thoughtful grouping mechanism, automate syntax and glossary maintenance, include few-shot examples and more selective retrieval could improve compatibility and reduce time and context costs. A glossary swarm spanning model companies would test construction-model bias; our non-Gemini experiments were constrained by personal funding. Future training should test broader data, reasoning-aligned objectives, and retention of existing abilities rather than assume that heavier fine-tuning will help. Training small models on English and BrainCode in parallel from scratch is another controlled comparison.

An additional direction would to look into Claude Haiku 4.5 surprising success rate compared to other and stronger models, and design experiments to figure whether these are circumstantial, unjustified confidence, or whether the model managed to learn a unique skill or state relevant to tasks as such. Independent creation runs could quantify structural and behavioral variance, testing whether model preferences recur across languages and strengthening both agent-capability and comparative-research conclusions. A further direction is explicit swarm communication: emergent communication research studies how agents acquire shared symbolic conventions (Havrylov and Titov, 2017; Lazaridou and Baroni, 2020). The reported Hugging Face incident illustrates the importance of observing and governing agent coordination (OpenAI, 2026). Auditable protocols warrant study; prevention of such incidents by BrainCode is untested.

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We’d like to thank researcher Guy Dar for his guidance and assistance with defining project’s goals and infrastructure. Also for providing us access to

Gemini models using his API key.

AI Disclosure

We used AI tools throughout the project as research and development assistants. Claude Opus 5.5 was used for assistance with code implementation from human instructions, pipeline design and refinements, adaptations to Guy’s swarm infrastructure, evaluation implementation, run monitoring, improvement suggestions, dataset retrieval, and design consultation. OpenAI Astra 6 also assisted with the initial specification and glossary refinements, comparison against Guy’s predefined syntax, and migration of its ideas into the syntax-loop output. Separate model agents used in the experiments are described in the main text.

The BrainCode swarm itself was operated using the researcher-provided API key and used Gemini 3.7 Flash during the relevant swarm runs. The swarm was used to process source items, generate BrainCode translations, and propose glossary additions or refinements as part of the language-construction process.

We also used AI during manuscript preparation. AI tools helped organize our outline, inspect repository code and saved experimental results, summarize design documents and qualitative reports, verify references, generate figures from stored results, and edit LaTeX and prose for clarity.

The reported experiments were not rerun for manuscript preparation. This disclosure describes assistance; it does not constitute an independent human validation of every semantic interpretation or generated translation.

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A Repository and Reproducibility

The project repository is <https://github.com/liasoffer/braincode-swarm>. The specification and glossary under `evaluations/runs/main/reference/` were compared with the current reference and found byte-identical. Core implementation and evidence locations are listed below; no claim is made that every local experimental artifact is already published remotely.

Models. The saved translation configuration identifies `vertex-proxy/gemini-flash` and `vertex-proxy/gemini-flash-high`, displayed as Gemini 3.7 Flash and its high-effort condition; `anthropic/claude-sonnet-5-5`; and `anthropic/claude-haiku-4-5-20251001`. Additional diagnostics use GPT-6.1 Sol, GPT-6 Luna, and o4-mini under the recorded harness configuration. These are experiment-time identifiers, not independently audited descriptions of proxy backends. The reasoning condition uses `meta-llama/Llama-3.1-8B-Instruct` through DeepInfra.

Dataset	Total	Local train	Local test
Mind2Web	2,022	1,617	405
ALFRED	10,001	7,975	2,026
SWE-bench	10,000	8,000	2,000
PRISM	8,011	6,408	1,603
PATHs	9,991	7,992	1,999
ThoughtTrace	2,155	1,725	430
Total	42,180	33,717	8,463

Table 4: Local ingestion and split counts. Each dataset contributes eight sampled evaluation items and three shared items; these counts do not describe official upstream benchmark splits.

Prepared training and statistical scope. The fine-tuning manifest contains 566 examples from 279 distinct items (535 training and 31 validation examples, with disjoint item identities), assembled from 232 swarm and 334 evaluation examples. Consequently any subsequent evaluation of a model trained on these data must use a fresh language-evaluation set. The Qwen checkpoint is evaluated on the separate ProofWriter benchmark (Appendix C). For the reasoning comparison, the saved run-level Wilson intervals are 7.7–18.2% (direct) and 0.1–3.7% (BrainCode), but these descriptive intervals do not account for item clustering; the paired and clustered analyses in Appendix C are more relevant to inference.

B Additional Determinism Views

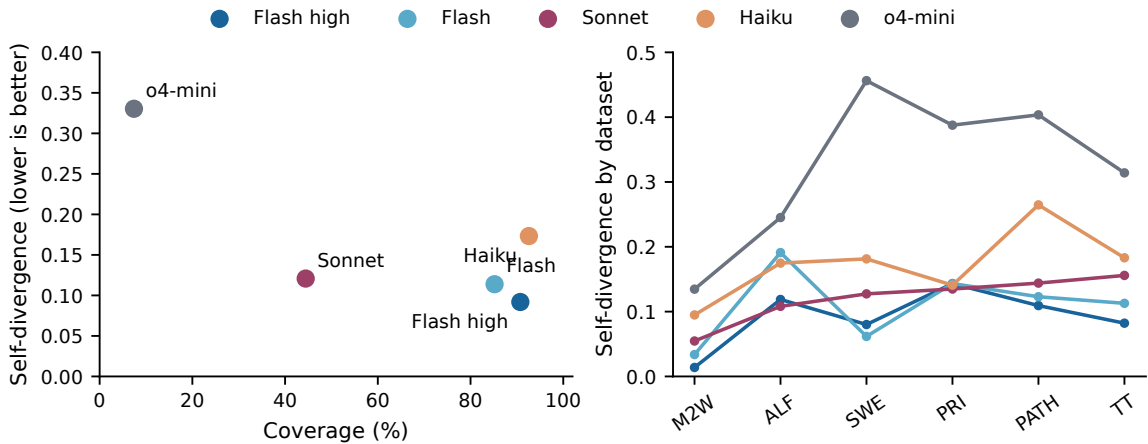


Figure 7: Shared-set coverage versus self-divergence (left) and self-divergence by dataset (right). o4-mini is a failure-confounded diagnostic. Dataset means use only items with at least two non-error translations; there are at most three items per dataset. Abbreviations follow Figure 6.

Figure 7 separates two objectives that a single success score obscures. Coverage ranks Haiku (92.6%), high-effort Flash (90.7%), Flash (85.2%), Sonnet (44.4%), and o4-mini (7.4%). Repeat stability, from lowest to highest divergence, ranks high-effort Flash (0.092), Flash (0.114), Sonnet (0.121), Haiku (0.173), and o4-mini (0.330). Per-dataset variation is substantial: Haiku’s largest divergence occurs on PATHs, while o4-mini’s largest occurs on SWE-bench. These small, selected subsets support descriptive comparisons only.

C Reasoning Experiments and Qwen Fine-Tuning

C.1 In-context Llama experiment on BBEH-mini

We use 50 items from BBEH-mini (Kazemi et al., 2025), stratified over all 23 task types: two per task plus four additional sampled items (seed 2026). Llama 3.1 8B Instruct (Grattafiori et al., 2024) completes three repeats per item and condition through the Hugging Face router/DeepInfra endpoint, with temperature 0.6, top- p 0.9, and at most 4,096 output tokens. The direct baseline receives the original task and a step-by-step instruction with the benchmark answer prefix. The treatment first translates with the specification and retrieved glossary, then solves in a fresh context from the translation plus language context. It does not retain the original task in the solving prompt.

Task	n	Direct	BrainCode
Boardgame QA	6	1 (16.7)	0 (0.0)
Boolean Expressions	6	0 (0.0)	0 (0.0)
Buggy Tables	6	0 (0.0)	0 (0.0)
Causal Understanding	6	1 (16.7)	0 (0.0)
Disambiguation QA	6	5 (83.3)	1 (16.7)
Dyck Languages	6	0 (0.0)	0 (0.0)
Geometric Shapes	6	0 (0.0)	0 (0.0)
Hyperbaton	6	0 (0.0)	0 (0.0)
Linguini	6	0 (0.0)	0 (0.0)
Movie Recommendation	6	1 (16.7)	0 (0.0)
Multistep Arithmetic	6	0 (0.0)	0 (0.0)
NYCC	6	2 (33.3)	0 (0.0)
Object Counting	6	0 (0.0)	0 (0.0)
Object Properties	6	0 (0.0)	0 (0.0)
SARC Triples	6	3 (50.0)	0 (0.0)
Shuffled Objects	9	2 (22.2)	0 (0.0)
Spatial Reasoning	6	0 (0.0)	0 (0.0)
SportQA	9	0 (0.0)	0 (0.0)
Temporal Sequence	9	0 (0.0)	0 (0.0)
Time Arithmetic	6	1 (16.7)	0 (0.0)
Web Of Lies	6	0 (0.0)	0 (0.0)
Word Sorting	9	1 (11.1)	0 (0.0)
Zebra Puzzles	6	1 (16.7)	0 (0.0)
Total	150	18 (12.0)	1 (0.7)

Table 5: BBEH-mini correct runs (accuracy in percent) by every sampled task type. n is the number of runs per condition, three per item. Scores use the benchmark evaluator; no runs errored.

Accuracy falls from 18/150 (12.0%) to 1/150 (0.7%), an 11.3-point decline (Table 5). The item-clustered GEE check gives $p = 0.0014$; a Bayesian random-intercept logistic model estimates an odds ratio of 0.05 (approximate 95% interval 0.01–0.19). However, majority voting within each item yields only five baseline-only successes and no treatment-only successes: exact paired McNemar $p = 0.0625$. The negative direction is consistent, while the strength of statistical evidence depends on aggregation and model assumptions.

Only 54% of treatment translations contain a recognized `braincode` block; otherwise the implementation forwards the entire translation response. Consequently this experiment tests the practical two-stage prompting pipeline, including its translation and formatting failures, not the intrinsic usefulness of perfectly formed BrainCode. Large language context, information loss, and lack of language-specific training are plausible explanations, but were not isolated by ablations. There is no matched-compute or alternative

structured-notation control. The original Llama fine-tuning attempt remained unevaluated; the later Qwen experiment below separately tests translation adaptation on ProofWriter.

C.2 Qwen translation fine-tuning

We adapted Qwen2.5-7B-Instruct (Qwen Team, 2025) to read and write BrainCode using 566 checked translations from 279 distinct source items. The item-disjoint split contains 535 training and 31 validation translations. Each produces an English-to-BrainCode example and a reverse BrainCode-to-English example; neither direction trains the model to solve the task. Filtering examples exceeding 8,192 tokens leaves 1,060 training and 62 validation sequences. ProofWriter is absent from this corpus; the original coverage/expressivity test items present in training must not be reused to evaluate the adapted model’s language generalization.

The saved Colab notebook records an A100 40GB run with 16-bit weights and LoRA. Rank and scaling are both 16, dropout is zero, and adapters cover attention projections (query, key, value, output) and MLP projections (gate, up, down), totaling 40,370,176 trainable parameters (0.53%). Response-only supervised training uses two epochs, learning rate 2×10^{-4} , cosine decay, 10 warmup steps, weight decay .01, 8-bit AdamW, effective batch size eight, and seed 2026. The run completes 266 optimizer steps; mean training loss is .561 and final validation loss .480. Translation loss is not a measure of reasoning ability.

C.3 ProofWriter design and completed comparisons

We sample 70 items from the depth-5 open-world test split of ProofWriter (Tafjord et al., 2021): 35 each with proof depth three and four, with True/False/Unknown counts 12/12/11 within each depth and one question per theory. Gemini 3.7 Flash translates 67/70 successfully (35/35 at depth three; 32/35 at depth four). The paired analysis uses these same 67 items in all Qwen conditions, with three repeats per item. English-input controls receive the original theory; BrainCode-input conditions receive its translation, the compact specification, and the glossary entries it uses. All conditions state the same open-world rule and request step-by-step reasoning followed by an English True/False/Unknown answer. Sampling uses temperature .7, top- p .8, top- k 20, repetition penalty 1.05, and a 4,096-token output limit.

The BrainCode conditions then make a second call to encode the English decision as a short BrainCode verdict. This call uses fixed templates, a prefixed block, greedy decoding, and a 400-token cap. A Gemini back-translator receives the final block and the question, but not the original theory. Because that decoding stage is incomplete in the saved report, Table 6 reports the completed first-call English answers. It does not present them as completed end-to-end BrainCode-verdict results. The report still awaits 201 fine-tuned and 83 base-model verdict reconstructions; the partially scored base-model verdict results are excluded from the paired table.

Model	Input	Correct / runs	Accuracy (%)	Change vs. base English
Qwen base	English	109/201	54.2	—
Qwen base	BrainCode	94/201	46.8	−7.5 points
Qwen fine-tuned	English	85/201	42.3	−11.9 points
Qwen fine-tuned	BrainCode	54/201	26.9	−27.4 points
Gemini Flash	English	65/67	97.0	—
Gemini Flash	BrainCode	61/67	91.0	—

Table 6: Completed ProofWriter answer comparisons on the 67 translated items. Qwen uses three repeats; Gemini is a single-run solvability reference with a larger output budget. All rows score English answers; changes use unrounded proportions.

For the combined fine-tuned/BrainCode condition versus base/English, per-item majority votes yield 21 baseline-only successes versus six treatment-only successes (exact McNemar $p = .0059$). The English-input fine-tuning control decreases by 11.9 points, but its majority-vote comparison is weaker ($p = .1338$), while the item-clustered GEE gives $p = .0076$. Base-model English versus BrainCode gives McNemar $p = .6177$. These tests answer different questions; the run-level decreases should not all be treated as

equally strong item-level evidence. On all 70 items, counting translation failures as wrong, base/English scores 54.8% and fine-tuned/BrainCode 25.7%.

The fine-tuned BrainCode condition predicts Unknown in 150/201 runs and produces 30 unparseable English answers, compared with 117 Unknown and no unparseable answers for base/English. This behavior is compatible with adaptation toward translation or restatement rather than logical inference. Small, heterogeneous translation data, lack of reasoning supervision, inherited translation errors, and changed answer-format behavior can all contribute. Catastrophic forgetting is a plausible risk, not a measured diagnosis: establishing general capability loss would require broader retained-task tests and training ablations. GPT-1 demonstrates successful transfer from generative pretraining (Radford et al., 2018); it does not establish that small-dataset fine-tuning necessarily harms a model. Work on ULMFiT directly motivates care about overfitting and forgetting (Howard and Ruder, 2018), although its setting differs from this LoRA experiment.

The Gemini solvability reference reaches 97.0% from English and 91.0% from BrainCode on the same 67 items. It uses one run per item and up to 16,000 output tokens, so it is not a matched model or compute control. Nevertheless, many translated problems remain solvable by a stronger reader, arguing against treating all of Qwen’s decline as irreversible loss in the representation alone.